Welfare, Integrability and Afriat

Advanced Microeconomics 1: Economic Decision Making. Lecture 4

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Where we left the question

The program so far, and the one gap left in it:

1. **L1**: the question stated; rationality testable; basket indexation over-compensates.
2. **L2**: the ideal answer defined — $e(p^1, u^0) - w$, per household.
3. **L3**: the answer made estimable — areas under Hicksian curves; bias $= \Delta p^{\mathsf{T}} S \Delta p / 2$.
4. **Today**: the missing license. Every bridge assumed a rational household *behind* the data. When does observed demand actually pin down preferences? Then: execute the answer on Belgium's index — and present the bill for every assumption used.

Invoice items 4 and 6: recovery, then audit.

Recovery I: demand functions (integrability)

The integrability problem

We know the forward direction: rational preferences $\Rightarrow x(p, w)$ satisfies Walras' law, h.d.0, and S symmetric $+$ NSD.

The reverse question (Sareh, §6)

Given a demand function $x(p, w)$ with those properties — does a rational preference relation exist that *generates* it?

Theorem · integrability (MWG 3.H)

Yes. If $x(p, w)$ satisfies Walras' law, h.d.0, and has a symmetric, negative semidefinite Slutsky matrix, then there exists a rational preference relation rationalizing x.

▸ proof

Proof: constructive sketch in the appendix; full treatment MWG 3.H. Sareh's notes state the result and, rightly, ask you to "realize the beauty of the theory."

Why this theorem is the license

Four consequences, in rising order of importance for us (Sareh's list, re-ordered):

1. Walras, h.d.0, symmetry, NSD are the *only* consequences of preference maximization for demand functions. The theory's complete fingerprint — four checkable conditions.
2. Symmetry is the *only* content beyond WARP's implications. Lecture 1's gap, measured exactly: one matrix condition.
3. An empirical strategy falls out: posit a tractable demand system, estimate it, *test* the restrictions (this is what Deaton–Muellbauer-style work does).
4. Welfare analysis is licensed: preferences can be *recovered* from observed demand — so e, CV, EV are computable from data. Without this theorem, Lectures 2–3 were a castle in the air. With it, the compensation program stands.

Recovery II: finite data (SARP and Afriat)

Finite data: strengthening WARP to SARP

An estimated demand *function* is a luxury. Often we hold finitely many observations — or 25 quiz choices.

Definition · SARP (Sareh, Def. 8.1; Houthakker 1950)

$x(p, w)$ satisfies **SARP** if for any chain $(p^1, w^1), \ldots, (p^N, w^N)$ with $x(p^{n+1}, w^{n+1}) \neq x(p^n, w^n)$ and $p^n \cdot x(p^{n+1}, w^{n+1}) \leq w^n$ for all $n \leq N - 1$, we have $p^N \cdot x(p^1, w^1) > w^N$.

Theorem · Houthakker (Sareh, Thm 8.1)

$x(p, w)$ satisfies SARP $\iff$ there is a rational preference relation that rationalizes x.

Proof deferred (Houthakker 1950; MWG 3.J). Chains may never cycle back — L1's three-good disease, outlawed at every length; equivalent to Slutsky symmetry + negativity. For $L = 2$: WARP $\iff$ SARP (MWG Ex. 3.J.1, a good self-test) — why the two-good quiz needed only the weak axiom.

Afriat's theorem: the code you ran last Thursday

Finite data $\{(p^t, x^t)\}_{t=1}^{T}$; GARP as in Lecture 1:

Theorem · Afriat (1967)

The following are equivalent:

(i) the data satisfy GARP;

(ii) there exist $U_t, \lambda_t > 0$ with $U_s \leq U_t + \lambda_t p^t \cdot (x^s - x^t)\ \forall s, t$ (*Afriat inequalities*);

(iii) the data are rationalizable by a continuous, increasing, **concave** utility function.

▸ proof

- For finite data, utility maximization *is* GARP. Concavity comes free: finite data can never reject it.
- Thursday, demystified: the GARP check was (i); the CCEI is the largest budget deflator at which (i) holds. [insert: GARP pass share; median CCEI; CKMS median.]

The answer, executed

Hausman's pipeline: one curve to the whole answer

Hausman (AER 1981), the block's (b)-methods paper, run on the thread:

1. **Estimate** one Marshallian demand: $x_E(p_E, w)$ for energy, from budget-survey data.
2. **Integrate**: Roy's identity, read backwards, is an ordinary differential equation in v; solve it along the price path. One estimated curve pins down indirect utility *exactly* — no functional leap of faith beyond the demand specification itself.
3. **Invert**: $e = v^{-1}$ in the wealth argument. Read off $e(p^1, u^0) - w$: the exact CV — and the deadweight loss of any tax on the good, exactly, not by triangle approximation.

Every step is a Lecture 3 bridge earning its keep: Roy backwards (step 2), duality (step 3), and integrability standing behind the whole thing — the license to treat the estimated x as *someone's* demand.

The Belgian index, audited

goal (b): policy

Hold the machinery up against the actual mechanism:

- **Fixed basket**: the health index is Laspeyres-type $\Rightarrow$ over-states the true rise by $\approx |\Delta p^{\mathsf{T}} S \Delta p|/2$. Measured through a surge: UK groceries 2021Q3–2023Q3, Laspeyres 26.2% vs. true cost-of-living 24.7% — 1.5pp of substitution in nine quarters (Chen–Levell–O'Connell 2025). [Boskin's US commission: total CPI bias $\sim$1.1pp/yr — verify before quoting.]
- **Exclusions**: the health index drops motor fuels, alcohol, tobacco. Whatever the rationale, excluding goods from the compensation index is *a welfare judgment* — it declares those price rises not owed to you. The theory makes the judgment visible; it cannot make it for you.
- **The central bank agrees**: NBB Annual Report 2022, Box 2 — published 2022 inflation "led to *overcompensation for the average household*." Lecture 1's three-line theorem, in an official document, with a second mechanism attached (new-contract energy pricing) that pushes the same way.

One number for everyone: who wins, who loses

goal (b): policy

The index is plutocratic (expenditure-weighted: each euro votes, so rich baskets count more) and unique (one π for all h). Against the machinery:

household	true cost change vs. index	indexation outcome
low income, no social tariff	$\pi^h > \pi$ (energy-heavy budget)	**under**-compensated
low income, social tariff	shielded on energy prices	closer to whole
average household	$\pi^h \approx \pi$, minus substitution	**over**-compensated (NBB)
high income	$\pi^h < \pi$; fast-indexing sectors	**over**-compensated
annual-January workers	correct π, one year late	real loss in the gap year
self-employed	no automatic indexation	bear the shock

Question (b), answered as far as theory plus official accounting can take it: indexation protects the average and misses the tails — in both directions. The tails are wide: in the UK surge, bottom-decile inflation ran 7.4pp above top-decile, and substitution offset less of the rise for the poorest quartile (5.5%) than the richest (6.9%) (Chen–Levell–O'Connell 2025).

The bill

What the euro answer leaned on

Every number this block produced rests on four load-bearing assumptions. Time to audit each:

1. **Rationality** — choices reveal a stable ranking. *Tested*: GARP, CCEI. On you.
2. **No money illusion** — h.d.0, filed away on the first content slide of Lecture 1.
3. **Stable preferences** — the u^0 being restored is the same object across the comparison.
4. **A choice of reference** — CV or EV; we treated it as bookkeeping.

None of these is safe. The next three slides present the evidence — starting with data you generated.

Exhibit A: your own answers

Week 1, two quiz questions, same object:

- "You own the mug — minimum price to sell?" [insert: class median WTA]
- "You don't — maximum price to buy?" [insert: class median WTP]
- Under assumptions 1–4, for a small-budget-share good, these should nearly coincide. [insert: class ratio.] The canonical finding: WTA $\approx 2\times$ WTP.

The block's third paper, revealed

Kahneman, Knetsch & Thaler (JPE 1990), "Experimental Tests of the Endowment Effect and the Coase Theorem" — mugs, real money, the gap survives markets and experience. Owning the reference point moves the valuation.

Consequence for us: CV and EV are elicited from different reference points — so they can differ *far* beyond wealth effects. "Which number does policy owe?" stops being bookkeeping.

Exhibit B: money illusion

Assumption 2 — h.d.0, “only relative prices and real wealth matter” — has its own file:

- Shafir, Diamond & Tversky (QJE 1997): in surveys and experiments, people systematically evaluate transactions partly in *nominal* terms — a 2% raise under 4% inflation feels better than a 2% cut under zero, same real outcome.
- Precisely the environment where it matters is ours: inflations. Wage bargaining, rent adjustment, the politics of indexation itself run partly on nominal frames — one reason indexation is beloved where it exists and resisted where it does not.
- Note the discipline: L1 *filed* the assumption instead of burying it; the theory told us exactly where to look for the crack. That is what “testable assumptions” buys.

The bill, itemized: if this fails, the answer errs that way

assumption at risk	evidence	direction of error in the com
substitution ignored (basket)	theorem (L1) $+$ NBB	over-pays; worst in relative
one index, many π^h	Jaravel; NBB quartiles	under-pays poor, over-pays
rationality (GARP fails, CCEI $<$ 1)	CKMS; [insert: this room]	recovery unlicensed for thos
no money illusion	SDT 1997	"real" compensation mis-pe
homotheticity (no trading down)	CLO 2025; EC1 P8	superlative indices track nei
reference independence (CV $=$ EV$+$w.e.)	KKT 1990	elicited answer depends on

- The point is not nihilism. Each row is a *signed, bounded* correction — the theory that produced the answer also produces its error analysis. That is the difference between a model and an opinion.

The block's scorecard

Takeaways

- **(a) Who did the 2022 shock hurt, and by how much?** Answered: π^h from shares alone; exact per-household CV via recovery + Hausman's pipeline; validity certified household-by-household by GARP/CCEI — including yours.
- **(b) Is one-basket indexation the right compensation?** No — it over-pays the average (theorem; NBB in print) and misses both tails (quartile weights; Jaravel). The right payment is $e^h(p^1, u^0) - w$, per household — and we can compute it.
- **The bill**: reference dependence, money illusion and failed rationality each bend the answer in a known direction. Use the numbers; carry the error bars.

Revealed reading: KKT (JPE 1990); Shafir, Diamond & Tversky (QJE 1997); Jaravel (QJE 2019) in full.

Ahead: EC2 (bring the practical output); then the firms block — where the prices that hurt households come *from*.

Appendix: proofs

Proof sketch: integrability (MWG 3.H)

Step 1 — recover e from x. Fix (p^0, w^0) and try to solve the system of PDEs given by Shephard run backwards:

$$\nabla_p\, e(p) \;=\; x\big(p,\, e(p)\big), \qquad e(p^0) = w^0.$$

A solution exists locally iff the system is consistent — Frobenius' condition: the "cross-derivatives" of the right-hand side must commute. Computing them, the condition is exactly symmetry of $S(p, e(p))$. That is the moment symmetry stops being decoration: it is the solvability condition for reconstructing the cost of living from demand.

Step 2 — recover preferences from e. Given $e(\cdot, u)$ (concave, by NSD, in the recovered system), define upper contour sets as the intersections of half-spaces

$$V_u \;=\; \{x \in \mathbb{R}^L_+ : p \cdot x \geq e(p, u) \;\text{ for all } p \gg 0\}.$$

Concave-duality (support-function) arguments show the preference relation these sets induce is rational and generates x. $\square$ (sketch; full argument MWG 3.H.1–3.H.2)

back

Proof sketch: Afriat's theorem

(iii) $\Rightarrow$ (i): a rationalizing utility orders the data; revealed-preference chains inherit the order, so no cycle can end strictly above its start.

(i) $\Rightarrow$ (ii): GARP makes the revealed relation acyclic, so observations can be ordered; a combinatorial induction then picks U_t, $\lambda_t > 0$ satisfying the inequalities — read as "a concave function with slope $\lambda_t p^t$ at x^t leaves room for U_s at x^s."

(ii) $\Rightarrow$ (iii): build the utility explicitly as the lower envelope of those supporting hyperplanes:

$$u(x) \;=\; \min_t \;\{\, U_t + \lambda_t\, p^t \cdot (x - x^t) \,\}$$

— continuous, increasing, concave by construction, and it rationalizes every observation.
$\square$

Sketch; Afriat (1967), Varian (1982). The CCEI relaxes (i): your Thursday number is the largest budget scaling $e^* \leq 1$ at which GARP holds.

back